

Course “Object-Oriented Programming in C++”**Week №13*****Homework*****Exercise 1.**

Create an abstract base class with a virtual function — the roots of the equation. Create derived classes: a class of linear equations and a class of quadratic equations. Determine the function calculating the roots of equations.

Exercise 2.

Create an abstract class Shape to draw plane figures. Define virtual methods:

- Show() — output to the screen the information about the figure,
- Save() — store the figure in the file,
- Load() — read the figure from the file.

Define derived classes:

- Square — square that is characterized by the coordinates of the upper left corner and the side length
- Rectangle — rectangle with the specified coordinates of the upper left corner and sizes
- Circle — circle with the specified center coordinates and radius
- Ellipse — ellipse with the specified coordinates of the upper angle of the rectangle that is described around this ellipse

with sides, which are parallel to axes, and with sizes of this rectangle.

Create an array of figures, save your figure to a file, upload to another array and display the information about each figure.